

PRACTICAL-1

AIM: Linux System Administration and File Management for Marketing Analytics Team

PROBLEM DEFINITION:

Scenario: You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers where project data, reports, scripts, and user accounts must be managed efficiently. Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team.

Tasks:

1. Identify the current user.
2. Find the current working directory.
3. Display system information.
4. Check system uptime and date.
5. Create a directory named MarketingAnalytics.
6. Create subdirectories: RawData, ProcessedData, Reports.
7. Create data and report files.
8. Add sample content.
9. Display file contents.
10. List all files and directories.
11. Search for report files.
12. Search specific content inside files.
13. Copy report files to Backup directory.
14. Rename files.
15. Move files between directories.
16. Delete a file.
17. Delete an empty directory.

SCREENSHOTS:

```
(kali㉿kali)-[~/Desktop]
$ whoami
kali

(kali㉿kali)-[~/Desktop]
$ pwd
/home/kali/Desktop

(kali㉿kali)-[~/Desktop]
$ uname -a
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux

(kali㉿kali)-[~/Desktop]
$ uptime
05:39:41 up 50 min,  5 users,  load average: 0.61, 0.90, 0.98

(kali㉿kali)-[~/Desktop]
$ date
Fri Jul 17 05:39:49 AM EDT 2026

(kali㉿kali)-[~/Desktop]
$ mkdir MarketingAnalytics

(kali㉿kali)-[~/Desktop]
$ cd MarketingAnalytics

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ mkdir RawData ProcessedData Reports

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ ls
ProcessedData  RawData  Reports

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ touch RawData/customer_data.txt

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ touch Reports/marketing_report.txt

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ echo "CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat" > RawData/customer_data.txt

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ echo "Marketing Analytics Report
Sales increased by 15%
Customer Engagement Improved" > Reports/marketing_report.txt

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ cat RawData/customer_data.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ cat Reports/marketing_report.txt
Marketing Analytics Report
Sales increased by 15%
Customer Engagement Improved

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ ls
ProcessedData  RawData  Reports

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ ls -l
total 12
drwxrwxr-x 2 kali kali 4096 Jul 17 05:42 ProcessedData
drwxrwxr-x 2 kali kali 4096 Jul 17 05:44 RawData
drwxrwxr-x 2 kali kali 4096 Jul 17 05:44 Reports

(kali㉿kali)-[~/Desktop/MarketingAnalytics]
$ ls -R
.:

```

```
ProcessedData RawData Reports
```

```
./ProcessedData:
```

```
./RawData:
```

```
customer_data.txt
```

```
./Reports:
```

```
marketing_report.txt
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ find . -name "*.txt"
```

```
./RawData/customer_data.txt
```

```
./Reports/marketing_report.txt
```

```
ProcessedData RawData Reports
```

```
./ProcessedData:
```

```
./RawData:
```

```
customer_data.txt
```

```
./Reports:
```

```
marketing_report.txt
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ find . -name "*.txt"
```

```
./RawData/customer_data.txt
```

```
./Reports/marketing_report.txt
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ find . -name "*report*"
```

```
./Reports/marketing_report.txt
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ grep "Raj" RawData/customer_data.txt
```

```
101,Raj,Ahmedabad
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ grep "Sales" Reports/marketing_report.txt
```

```
Sales increased by 15%
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ mkdir Backup
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ cp Reports/marketing_report.txt Backup/
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ ls
```

```
Backup ProcessedData RawData Reports
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ mv Reports/marketing_report.txt Reports/final_report.txt
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ mv RawData/customer_data.txt ProcessedData/
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ rm Backup/marketing_report.txt
```

```
(kali㉿kali)-[~/Desktop/MarketingAnalytics]
```

```
$ rmdir Backup
```

KEY QUESTIONS / ANALYSIS:

- 1) What is the difference between absolute and relative paths?
 - Absolute path: Specifies the complete location of a file or directory starting from the root (/).
Example: /home/user/MarketingAnalytics/Reports/report.txt
 - Relative path: Specifies the location relative to the current working directory. Example:
Reports/report.txt
- 2) Why are Linux permissions important for file security?

Linux permissions control who can read (r), write (w), and execute (x) files and directories. They prevent unauthorized access, protect sensitive data, and help maintain system security.
- 3) What is the purpose of the grep command?

The grep command is used to search for specific words or patterns within files. It helps users quickly find relevant information in text files.
- 4) How does Linux support efficient file management?

Linux provides a hierarchical file system and powerful commands such as ls, cp, mv, rm, find, and grep for organizing, searching, copying, moving, and managing files efficiently.
- 5) Why are backups important in enterprise environments?

Backups protect data from accidental deletion, hardware failures, cyberattacks, and system crashes. They ensure business continuity and enable quick recovery of important files and systems.

CONCLUSION:

In this practical, Linux system administration and file management tasks were successfully performed using basic Linux commands. Directories and files were created, modified, searched, copied, moved, renamed, and deleted efficiently. System information, user details, and resource status were also checked. This practical improved understanding of Linux file management, permissions, and backup concepts, demonstrating how Linux provides a secure and efficient environment for managing enterprise data.